Representation Multiplicity in Causal Forests

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Abstract

Feature-subsampled forests treat split-equivalent covariate columns as distinct candidates. Keeping a covariate together with strict monotone transformations therefore changes how often the covariate is considered without changing available information. For an honest causal forest satisfying continuation and score-transfer conditions, I show that path depth times singleton inclusion probability governs the limiting CATE target. When this index vanishes, two information-equivalent representations can disagree in sign on half the population and induce different treatment rules. When it converges to a finite positive constant, the singleton component is attenuated. When it diverges, both forests recover the CATE. The default candidate rule yields a squared-depth multiplicity boundary. Monte Carlo and an Econometrica replication show finite-sample sensitivity. Sampling certified split classes restores exact invariance on a declared array.

Keywords: causal forests, feature subsampling, policy learning, representation invariance, redundant covariates.

JEL Classification: C14, C21, C55

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1 Introduction

Suppose two researchers use a causal forest to estimate how a job-training program affects later earnings as a function of baseline household income and age. One includes income and age. The other keeps age but includes income, log income, and several other strictly increasing recodings of income. If baseline income has positive support, these income columns generate the same threshold partitions and contain the same covariate information. A feature-subsampled forest nevertheless treats them as separate candidates, making an income split more likely to enter the node-level search and an age split less likely.

The effect of this change depends on the whole path to a leaf. A covariate that drives treatment-effect heterogeneity matters only after a split separates its relevant states. If such a covariate is represented once among many columns, each node offers a small chance to use it. Those chances accumulate with path depth. This paper studies the resulting access-depth interaction.

The main result uses two independent covariates with binary treatment-effect components. One has a smaller coefficient than the other. The first representation repeats the covariate with the smaller coefficient and leaves the other as a singleton. The second does the reverse. Both representations generate the same sigma-field and the same split-action library. Let H_n be effective path depth, M_n the number of columns, and Q_{M_n} the number sampled at a node. The relevant index is $\mathcal{I}_n = H_n\mathbb{E}[Q_{M_n}]/M_n$.

Under the continuation and score-transfer conditions stated below, if $\mathcal{I}_n \to 0$, the singleton covariate remains unused along a prediction path with probability approaching one. Each forest then retains only its repeated component. The two CATE limits have opposite signs on half the covariate population, and repeating the covariate with the smaller coefficient produces nonvanishing policy regret. If $\mathcal{I}_n \to \kappa \in (0, \infty)$, the singleton component enters with a strictly intermediate weight under the additional path-depth condition in Theorem 1. If $\mathcal{I}_n \to \infty$, both forests recover the CATE. Every fixed finite representation covered by the theorem therefore recovers under the stated conditions. The failure is a lack of uniformity over growing, information-equivalent representations.

The candidate-count rule gives a transparent boundary. In the `grf` R package for generalized random forests, the `mtry` parameter controls how many covariate columns are randomly sampled as split candidates at each node. With `mtry` fixed, $\mathcal{I}_n$ has order H_n/M_n. Under the package-default square-root scaling in `grf`, it has order $H_n/\sqrt{M_n}$. The critical representation size is thus of order H_n in the first case and H_n^2 in the second.

With 500 baseline columns and eight equivalent recodings, the Monte Carlo forests disagree in CATE sign for 45.7 percent of evaluation observations. Changing only the forest seed gives 0.09 percent. Class sampling eliminates representation-induced disagreement and reduces policy regret in this design, although its CATE mean squared error is higher. The random-forest outcome-regression specifications in Chernozhukov et al. [2022] retain age, education, and two prior-earnings variables together with their squares. Grouping columns that rank the observations identically and retaining one representative from each group substantially changes some individual first-stage contrasts. The six reconstructed ATETs change only modestly, remaining positive and yielding the same conclusions about statistical significance at the 5 percent level.

The remedy is to sample split-action classes rather than raw column labels. On a declared finite array, columns with the same weak ordering and ties generate the same threshold actions. Collapsing them before candidate sampling gives exact equality of tree partitions, forest weights, predictions, reported standard errors, and selected sets under common class-level randomness. In the application of Chernozhukov et al. [2022], this removes the extra candidate weight attached to a polynomial label without discarding any certified CART threshold partition.

Louppe [2014] explains how redundant predictors change randomized split access, tree structure, and prediction. Related work studies variable selection and importance [Strobl et al., 2007, Toloşi and Lengauer, 2011, Gregorutti et al., 2017, Kubuś, 2018]. Recovery bounds already relate candidate access to depth [Chi et al., 2022, Klusowski and Tian, 2024], and Mei et al. [2026b] explicitly characterize unresolved-component bias for population forests with training-data-independent partitions. Mei et al. [2026a] study sequential allocation of split opportunities. Here the partitions are grown from the sample and stopping is data dependent. Under the stated continuation and score-transfer conditions, the theorem derives their representation-specific prediction limits, with intermediate attenuation conditional on the failure-path depth limit. Sign disagreement and regret follow from these limits in the omission regime. The square-root-budget boundary specializes this analysis to a familiar recovery scale. Module sampling [Chen and Zhang, 2013] and inverse-cluster-size feature weighting [Colot et al., 2021] are antecedents of the correction. The class-sampling result provides an exact invariance guarantee on a certified finite array. Related work on honest forests and treatment rules includes Wager and Athey [2018], Athey et al. [2019], Cattaneo et al. [2022], Cattaneo et al. [2025], Hirano and Porter [2009], Kitagawa and Tetenov [2018], and Athey and Wager [2021].

Section 2 defines the model. Section 3 derives the forest limits and their treatment-rule consequences. Section 4 gives the class-sampling guarantee, and Section 5 reports the evidence. All proofs are in the appendix.

2 Model Setup

Let $X = (X_1, X_2) \sim \mathrm{Unif}([0,1]^2)$ be the covariate vector. For coordinate $j \in \{1,2\}$, define the binary state indicator

$$s_j(X) = 2\mathbf{1}\{X_j \geq 1/2\} - 1,$$

and define the conditional average treatment effect (CATE)

$$\tau_a(X) = a_1 s_1(X) + a_2 s_2(X),$$

indexed by the coefficient vector $a = (a_1, a_2)$. Restrict a to the parameter class

$$\mathcal{C}_a = \{(a_1, a_2) : \underline{a} \leq a_1 < a_2 \leq \overline{a},\ a_2 - a_1 \geq \Delta_a\}$$

where $\underline{a}$ and $\overline{a}$ are fixed lower and upper coefficient bounds satisfying $0 < \underline{a} < \overline{a}$, and Δ_a is a fixed minimum coefficient gap, with $0 < \Delta_a < \overline{a} - \underline{a}$.

Let n be the sample size, and let $(Y_i, W_i, X_i)_{i=1}^n$ denote the i.i.d. observations. For treatment state $w \in \{0,1\}$, the potential outcome $Y(w)$ follows

$$Y(w) = m_0(X) + w\tau_a(X) + U, \qquad \mathbb{E}(U \mid X, W) = 0, \tag{1}$$

where m_0 is the baseline outcome function and U is the outcome disturbance. The binary treatment indicator W is randomized with propensity score one half, and the observed outcome Y satisfies

$$W \perp \{Y(0), Y(1)\} \mid X, \qquad \mathbb{P}(W = 1 \mid X) = \frac{1}{2}, \qquad Y = Y(W).$$

The baseline function m_0 and the conditional law of U are held fixed over $a \in \mathcal{C}_a$. Define the observed-outcome regression $m_a(X) = \mathbb{E}(Y \mid X) = m_0(X) + \tau_a(X)/2$. The oracle transformed-outcome score for observation i is

$$\Gamma_i = 4(W_i - 1/2)\{Y_i - m_a(X_i)\} \tag{2}$$

and satisfies $\mathbb{E}(\Gamma_i \mid X_i = x) = \tau_a(x)$ at every covariate value x. The fitted score $\widehat{\Gamma}_i$ replaces m_a with the cross-fitted outcome-regression estimator $\widehat{m}$.

For covariate index $j \in \{1,2\}$, let $c_n \geq 1$ be the representation multiplicity. Choose strictly increasing recoding maps $\{\varphi_{j\ell n} : \ell = 1, \ldots, c_n\}$, including the identity, and denote the ℓth recoded column by $X_j^{(\ell)} = \varphi_{j\ell n}(X_j)$. Define the paired covariate representations

$$R_{1,n}(X) = \{X_1^{(1)}, \dots, X_1^{(c_n)}, X_2\}, \qquad R_{2,n}(X) = \{X_1, X_2^{(1)}, \dots, X_2^{(c_n)}\}.$$

For representation index $h \in \{1, 2\}$, $R_{h,n}$ denotes representation h, and $M_n = c_n + 1$ is its raw column count. Both representations generate the covariate sigma-field $\sigma(X_1, X_2)$. A cut $X_j \leq t$ corresponds to $\varphi_{j\ell n}(X_j) \leq \varphi_{j\ell n}(t)$ in each recoded column. These corresponding cuts preserve partitions and routing in exact arithmetic. The first representation repeats X_1, whose coefficient is smaller. The second repeats X_2, whose coefficient is larger.

Each tree draws s_n observations uniformly without replacement and randomly divides them into splitting and estimation samples, independently of their values. Both sample fractions are at least a fixed $\eta \in (0, 1/2)$. Write n_C for the splitting-sample count in node C. At a node with M raw columns, the tree draws $Q_M \in \{1, \dots, M\}$ and samples that many columns uniformly without replacement, independently of the data and previous candidate draws. It evaluates every distinct threshold partition of the sampled columns. A split is admissible when each child contains at least k_n splitting observations and at least a fraction $\beta \in (0, 1/2)$ of the parent's splitting sample. The tree chooses a maximum-gain admissible split, using a representation-independent order for exact ties. It stops if no sampled admissible action has positive gain, with no separate depth cap or stopping rule. The estimation sample supplies the mean fitted score in each leaf, with value zero assigned to an empty estimation leaf. Conditional on the training data, fitted nuisance functions, and root sample assignments, each node's action depends only on its ancestral states, its observations, and fresh randomness assigned to that node. Randomization is independent across node addresses. Actions do not depend on candidate draws or split decisions on other branches.

A coordinate is used on a prediction path once it has been chosen as the splitting coordinate. For a generic tree under representation h, let $D_{h,n}(x)$ indicate that the singleton X_{3-h} has been used before the path to x stops. First use need not make the leaf exactly pure in s_{3-h}. The proof controls the resulting approximation error in $L^1(P_X)$, where P_X is the covariate distribution. For a candidate split indexed by column j and threshold t, let C be the parent cell and C_L and C_R its left and right child cells. Define the pseudo-outcome split-gain criterion

$$\widehat{\mathcal{Q}}_C^{\Gamma}(j, t) = \widehat{p}_{L|C} \widehat{p}_{R|C} \left(\overline{\widehat{\Gamma}}_{C_L} - \overline{\widehat{\Gamma}}_{C_R} \right)^2. \tag{3}$$

Here $\widehat{p}_{L|C}$ and $\widehat{p}_{R|C}$ are the empirical child shares within C, and $\overline{\widehat{\Gamma}}_{C_L}$ and $\overline{\widehat{\Gamma}}_{C_R}$ are the corresponding child-cell score means. Let $\widehat{\mathcal{Q}}_C^{L}(j, t)$ be the forest algorithm's normalized split-gain criterion, $\check{T}_{h,n}$ the arithmetic mean of the B_n honest score-leaf predictions, and $\widehat{T}_{h,n}$ the reported forest predictor under representation $R_{h,n}$. The following conditions connect the forest algorithm to the score and control tree depth.

Assumption 1. *All probability statements and stochastic rates in this assumption hold uniformly over $a \in \mathcal{C}_a$. Outcomes are uniformly bounded, the conditional law of U has a continuous density on bounded support, and $\|\widehat{m} - m_a\|_\infty = o_p(1)$, with $\|\widehat{m}\|_\infty = O_p(1)$, where $\|\cdot\|_\infty$ is the sup norm. For a fixed $C_0 > 4$, with probability tending to one, every representation column has a positive-gain admissible threshold at every reached node with $n_C \geq C_0 k_n$, on ordinary trees and on the locally conditioned failure trees defined below, whenever those trees are defined. Let $\bar{s} < 1$ be a fixed upper bound on the tree subsampling fraction. Then*

$$\begin{aligned} s_n \to \infty, \qquad & s_n/n \leq \bar{s} < 1, \qquad k_n \to \infty, \\ k_n/s_n \to 0, \qquad & \frac{k_n}{\log(B_n s_n)} \to \infty, \qquad B_n \to \infty. \end{aligned} \tag{4}$$

On the actual and potential failure trees,

$$\sup_{C,j,t} \left| \widehat{\mathcal{Q}}_C^L(j,t) - \widehat{\mathcal{Q}}_C^\Gamma(j,t) \right| = o_p(1), \qquad \max_{h=1,2} \|\widehat{T}_{h,n} - \check{T}_{h,n}\|_{L^1(P_X)} = o_p(1). \tag{5}$$

Conditional on the training data and fitted nuisance functions, tree groups are independent, have uniformly bounded size, and have the same marginal tree distribution.

The first condition in (5), the criterion transfer, requires uniformly close gain values and hence preserves comparisons separated by a fixed positive gap. The second, the prediction transfer, requires its reported predictor to agree with the honest score-leaf forest. These algorithm-specific conditions are combined with the candidate-count law. The continuation condition ensures growth to the minimum-size region even when only a previously used coordinate is offered. Balance and vanishing criterion error alone do not ensure this. The theorem requires these conditions to be checked for the forest algorithm under study.

Finally, define the effective path depth H_n, the singleton's per-node inclusion probability ω_n, and the access-depth index $\mathcal{I}_n$ by

$$H_n = \log(s_n/k_n), \qquad \omega_n = \frac{\mathbb{E}Q_{M_n}}{M_n}, \qquad \mathcal{I}_n = H_n \omega_n. \tag{6}$$

The access-depth index combines the number of opportunities along a path with the chance of accessing the singleton at each node.

3 Forest Limits and Treatment-Rule Distortion

For $\omega_n < 1$, define a failure tree by conditioning the ordinary transition at each visited node locally on not choosing the singleton as the splitting coordinate. A conditioned transition can split on the repeated coordinate or stop. Its conditioning probability is at least $1 - \omega_n > 0$. Let $J^0_{h,n}(X)$ count its candidate-set draws at size-eligible nodes, including a final unsuccessful search if one occurs. In the intermediate regime $\mathcal{I}_n \to \kappa \in (0, \infty)$, $\omega_n \to 0$, so this construction is eventually defined. For each fixed $a \in \mathcal{C}_a$, additionally suppose

$$\frac{J^0_{h,n}(X)}{H_n} \longrightarrow_p d_h(a) \in (0, \infty), \qquad h = 1, 2, \tag{7}$$

where the convergence accounts for randomness in the training sample, the locally conditioned failure-tree construction, and an independent test point $X \sim P_X$.

Theorem 1 (Representation-dependent forest limits)**.** *Under Assumption 1, the two endpoint results hold uniformly over $a \in \mathcal{C}_a$. If $\mathcal{I}_n \to 0$, then*

$$\max_{h=1,2} \|\widehat{T}_{h,n} - a_h s_h\|_{L^1(P_X)} \longrightarrow_p 0. \tag{8}$$

If $\mathcal{I}_n \to \infty$, then

$$\max_{h=1,2} \|\widehat{T}_{h,n} - \tau_a\|_{L^1(P_X)} \longrightarrow_p 0. \tag{9}$$

For fixed $a \in \mathcal{C}_a$, if $\mathcal{I}_n \to \kappa \in (0, \infty)$ and (7) holds, then

$$\begin{aligned} \widehat{T}_{1,n} &\longrightarrow_p a_1 s_1 + \{1 - e^{-\kappa d_1(a)}\} a_2 s_2, \\ \widehat{T}_{2,n} &\longrightarrow_p \{1 - e^{-\kappa d_2(a)}\} a_1 s_1 + a_2 s_2 \end{aligned} \qquad \textit{in } L^1(P_X). \tag{10}$$

The index $\mathcal{I}_n$ measures how many opportunities the singleton has along a path. When the index vanishes, the singleton remains unused with probability approaching one, and its component drops out of the limit. In the intermediate regime, $e^{-\kappa d_h(a)}$ is the limiting probability that a path reaches its leaf without using the singleton, so the singleton component is attenuated by the complementary factor $1 - e^{-\kappa d_h(a)} \in (0, 1)$. When the index diverges, both components are recovered in the forest prediction. The result applies equally to exact copies and to growing sequences of strictly increasing recodings.

For a treatment rule π, let

$$V_a(\pi) = \mathbb{E}\{\tau_a(X)\pi(X)\}, \qquad \mathrm{Reg}_a(\pi) = V_a(\pi^*_a) - V_a(\pi),$$

The optimal and fitted rules are

$$\pi_a^*(x) = \mathbf{1}\{\tau_a(x) > 0\} = \mathbf{1}\{s_2(x) = 1\}, \qquad \widehat{\pi}_{h,n}(x) = \mathbf{1}\{\widehat{T}_{h,n}(x) > 0\}.$$

Corollary 1 (Sign reversal and policy regret)**.** *Under the conditions of Theorem 1, if $\mathcal{I}_n \to 0$, then*

$$\mathbb{P}_X\{\widehat{T}_{1,n}(X)\widehat{T}_{2,n}(X) < 0\} \longrightarrow_p \frac{1}{2}. \tag{11}$$

The disagreement remains separated from zero, and

$$\mathrm{Reg}_a(\widehat{\pi}_{2,n}) \longrightarrow_p 0, \qquad \mathrm{Reg}_a(\widehat{\pi}_{1,n}) \longrightarrow_p \frac{a_2 - a_1}{2}. \tag{12}$$

Thus the representation that repeats the covariate with the larger coefficient yields the optimal unconstrained rule in the selection regime. Repeating the covariate with the smaller coefficient yields a persistent treatment mistake. The conclusion concerns an unconstrained rule. Under a capacity constraint, representation changes act through ranks near the assignment cutoff. To connect these asymptotic regimes to the forest's tuning parameters, consider the candidate-count law. In `grf` 2.4.0,

$$N_M \sim \mathrm{Poisson}(\xi_M), \qquad Q_M = \max\{1, \min(N_M, M)\}, \qquad \xi_M = \texttt{mtry} \tag{13}$$

[GRF Development Team, 2026].

Corollary 2 (Multiplicity boundary)**.** *Under the conditions of Theorem 1, if $M_n \to \infty$ and $\xi_{M_n} \equiv \xi > 0$, then*

$$\omega_n \sim \frac{\xi + e^{-\xi}}{M_n},$$

so the critical representation size is $M_n \asymp H_n$. Under the package default $\xi_M = \min\{\lceil \sqrt{M} \rceil + 20, M\}$, $\omega_n \sim M_n^{-1/2}$, and the critical representation size is $M_n \asymp H_n^2$. Every fixed finite representation satisfies $\mathcal{I}_n \to \infty$.

The theorem does not invalidate fixed-dimensional causal-forest consistency. Instead, it adds a rate requirement. Recovery requires the access-depth index $H_n \mathbb{E}(Q_{M_n})/M_n$ to diverge. Every fixed finite representation satisfies this condition asymptotically, but an empirical forest is fitted at a finite sample size, where the index remains finite and may be small. Representation sensitivity can therefore remain practically important even when asymptotic consistency eventually holds.

4 Finite-Array Class Sampling

An invariant forest can sample split classes instead of raw columns. Fix an array $\mathcal{X}_{\text{cert}} = \{x_1, \dots, x_N\}$ containing every point used for fitting, prediction, variance estimation, and policy ranking. A column's certificate is its weak ordering of this array, identified with the reversed ordering. Equal certificates generate the same unordered threshold partitions on every subset of the array.

Construct the classes before growing the forest. Replace each raw column on $\mathcal{X}_{\text{cert}}$ by its weak-rank vector, assigning the same rank to tied observations, and treat a rank vector and its reversal as the same certificate. Group columns with identical certificates, choose a canonical orientation and weak-rank representative for each group, and order the groups by their certificates. At each node, sample these groups uniformly without replacement and evaluate the usual CART threshold splits using their canonical representatives. Index child orientation, tie-breaking, folds, and all other randomization by the certificate rather than by the original column label. Hold nuisance fits fixed across representations or fit them on the same canonical matrix. All tuning rules and reported outputs use this matrix and the same settings. Because this reduction occurs before candidate sampling, adding or removing a split-equivalent recoding does not alter either the candidate probabilities or the random draws.

Proposition 1 (Class-sampled forest)**.** *Fix the certified classes, canonical routing, and class-level candidate law. Under common class-indexed randomness, permutation, insertion, or deletion within a represented class, complementation, and every recoding that preserves or reverses the observed weak ordering and ties leave all tree partitions, forest weights, predictions, reported standard errors, and selected sets on $\mathcal{X}_{\text{cert}}$ unchanged.*

This guarantee is transductive. It covers the declared array, not future points whose ordering has not been certified, and it does not merge correlated variables with different orderings.

5 Finite-Sample Evidence

5.1 Monte Carlo

The Monte Carlo independently draws $X \sim \text{Unif}([0,1]^p)$, $W \sim \text{Bernoulli}(1/2)$, and

$$\begin{aligned}
\tau(X) &= 0.20 + 0.65 s_1 - 0.55 s_2 + 0.10(X_3 - 0.5), \\
Y &= 1 + X_4 + 0.5 X_5^2 - 0.5 X_6 + W\tau(X) + \varepsilon, \qquad \varepsilon \sim N(0,1).
\end{aligned}$$

Table 1 compares paired representations that add eight monotone or order-reversing recodings of X_1 or X_2. These are standardization, complementation, an affine change of units, a shifted logarithm, ranks, a shifted square, an exponential, and a shifted cube. Each cell averages 100 independent samples of 3,000 observations, with 30,000 evaluation observations and 2,000-tree forests fitted with `grf` 2.4.0. Both representations use the true nuisance functions $\mathbb{E}(Y \mid X)$ and $\mathbb{P}(W = 1 \mid X) = 1/2$. Thus the comparison isolates the forest fit from nuisance-estimation error. The forests use default `mtry`, a minimum node size of five, and honest half-samples with an equal split between partition construction and leaf estimation. Data and forest seeds are paired. For predictions $\widehat{\tau}_1$ and $\widehat{\tau}_2$, SR_c is the fraction with opposite signs and both magnitudes at least c.

Table 1: CATE sign disagreement under equivalent representations

Comparison	Raw columns	SR_0 (%)	$\mathrm{SR}_{0.10}$ (%)
$p = 40$, representation change	48	8.81 (0.66)	0.02 (0.003)
$p = 100$, representation change	108	23.41 (0.66)	12.70 (0.83)
$p = 250$, representation change	258	38.60 (0.84)	26.87 (0.73)
$p = 500$, representation change	508	45.74 (0.54)	32.88 (0.91)
$p = 500$, second forest seed	508	0.09 (0.01)	0.00 (0.00)
$p = 500$, class-sampled forest	508	0.00 (0.00)	0.00 (0.00)

Notes: Entries are means across 100 independent replications, with Monte Carlo standard errors in parentheses. The first four rows compare representations with eight added encodings. The last two rows change only the seed or apply class sampling to the $p = 500$ design.

Sign disagreement rises sharply with the ambient number of columns. At $p = 500$, 32.88 percent of observations have opposite signs with both estimates at least 0.10 in magnitude. Changing only the forest seed produces 0.09 percent sign disagreement. Class sampling removes the difference exactly on the pooled training and evaluation array.

Accuracy and treatment decisions need not move together. For evaluation observations $X_1^e, \ldots, X_N^e$, with $N = 30{,}000$, define

$$\mathrm{MSE}_j = \frac{1}{N} \sum_{i=1}^{N} \{\widehat{\tau}_j(X_i^e) - \tau(X_i^e)\}^2,$$

$$\mathrm{Reg}_j = \frac{1}{N} \sum_{i=1}^{N} |\tau(X_i^e)| \mathbf{1}\{\widehat{\pi}_j(X_i^e) \neq \pi^*(X_i^e)\},$$

where j indexes the fitted forest, $\widehat{\pi}_j(x) = \mathbf{1}\{\widehat{\tau}_j(x) > 0\}$, and $\pi^*(x) = \mathbf{1}\{\tau(x) > 0\}$. Regret measures forgone treatment gains relative to this optimal unconstrained rule. Table 2 reports both losses and paired differences for class sampling, which was evaluated at $p = 500$. The class-sampled fits use the same oracle nuisance functions, certify weak-order classes on the pooled training and evaluation covariates, and apply the default candidate budget to the number of classes.

Table 2: CATE accuracy and treatment-rule regret

p	Forest	MSE	Regret
40	Recodings of X_1	0.0361 (0.0012)	0.0127 (0.0009)
	Recodings of X_2	0.0296 (0.0010)	0.0058 (0.0007)
100	Recodings of X_1	0.0747 (0.0018)	0.0222 (0.0006)
	Recodings of X_2	0.0615 (0.0018)	0.0093 (0.0015)
250	Recodings of X_1	0.1450 (0.0019)	0.0257 (0.0001)
	Recodings of X_2	0.1511 (0.0023)	0.0438 (0.0026)
500	Recodings of X_1	0.1962 (0.0015)	0.0262 (0.0001)
	Recodings of X_2	0.2346 (0.0020)	0.0647 (0.0016)
	Class sampling	0.3653 (0.0021)	0.0047 (0.0004)
Paired differences at $p = 500$			
Class sampling minus X_1 recodings		0.1690 (0.0014)	-0.0215 (0.0004)
Class sampling minus X_2 recodings		0.1307 (0.0012)	-0.0600 (0.0017)

Notes: Means across the same 100 replications as the representation comparisons in Table 1. Parentheses give Monte Carlo standard errors. For paired differences, these are the sample standard deviation of the within-replication differences divided by $\sqrt{100}$. Both class-sampled representations have exactly identical predictions in every replication. Regret is in outcome units per person.

At $p = 500$, class sampling has mean squared error 0.3653, compared with 0.1962 and 0.2346 when recodings of X_1 and X_2 are retained. Its mean regret is 0.0047, compared with 0.0262 and 0.0647. Class sampling improves treatment decisions in this design while increasing mean squared CATE error. A sign-based rule depends on which side of the treatment cutoff an estimate falls. Squared error also penalizes mistakes in effect magnitudes.

5.2 A published random-forest first stage

The NSW application in Chernozhukov et al. [2022] includes age, education, 1974 earnings, and 1975 earnings together with their squares. Since their observed supports are nonnegative, each raw-square pair offers the same CART threshold partitions. I certify classes on the pooled observed and treatment-counterfactual rows, including both treatment states for every covariate vector used in prediction. The reduction retains one original column per class and removes four or five of 24 to 30 columns. It changes only the random-forest outcome regression, leaving the samples, folds, Riesz/Lasso stage, and orthogonal score fixed. The native comparison uses original representative values. A second comparison gives equivalent columns identical canonical ranks before either dictionary is fitted. The six cells combine the NSW experimental comparison and the PSID and CPS comparison groups with specifications 1 and 2 of the original paper. The reconstructed full-dictionary ATETs are within 0.10 published standard errors of the reported estimates.

Table 3 reports both reductions. One recomputes the `randomForest` regression default, one third of the supplied column count rounded down, after removing columns. The other holds its original numerical value fixed. All fits use 1,000 trees and the same seed. The final column benchmarks representation changes against four alternative seeds of the full dictionary.

Table 3: ATET estimates with full and reduced forest dictionaries

Sample / specification	Full dictionary	Reduced, default `mtry`	Reduced, fixed `mtry`
NSW / 1	3,013.6 (1,277.9)	3,127.8 (1,322.0)	3,043.6 (1,304.9)
NSW / 2	3,207.0 (1,364.9)	3,191.1 (1,359.7)	3,038.1 (1,264.3)
PSID / 1	1,469.0 (975.6)	1,555.0 (924.2)	1,645.8 (941.6)
PSID / 2	1,381.1 (949.0)	1,362.6 (921.7)	1,345.0 (931.0)
CPS / 1	1,636.2 (616.3)	1,483.6 (599.7)	1,621.5 (607.5)
CPS / 2	1,566.2 (618.3)	1,505.1 (605.7)	1,570.0 (608.5)
	Change from full dictionary		Seed variation
	Default `mtry`	Fixed `mtry`	Mean absolute (max.)
NSW / 1	+114.3	+30.0	43.8 (86.4)
NSW / 2	-15.9	-169.0	176.1 (234.7)
PSID / 1	+86.0	+176.8	36.4 (61.4)
PSID / 2	-18.5	-36.1	46.1 (64.8)
CPS / 1	-152.6	-14.7	22.3 (30.4)
CPS / 2	-61.1	+3.8	10.1 (20.0)

Notes: All amounts are in dollars. Upper-panel parentheses give ATET standard errors. Full-dictionary entries are reconstructed estimates, not the published table entries. Each fit uses 1,000 trees and fixed five-fold cross-fitting. Lower-panel changes pair seed 1 before and after reduction. Seed variation compares the full-dictionary estimate at seed 1 with seeds 2–5. Specifications 1 and 2 have 24 and 30 original columns. Reduction leaves 19 and 25 columns in NSW and CPS, and 20 and 26 in PSID. Recomputed `mtry` is 6 and 8, versus 8 and 10 when held fixed.

The ATET changes range from 4 to 177 dollars in absolute value. Every estimate remains positive, and every 95 percent significance classification is unchanged. Representation changes do not uniformly exceed seed variation in the aggregate ATET. For example, in NSW specification 2, the default-budget reduction changes the estimate by 16 dollars, compared with a mean absolute seed change of 176 dollars.

For the cross-fitted outcome regression $\widehat{m}(w, x)$, the sign of $\widehat{m}(1, x) - \widehat{m}(0, x)$ changes for 2.3–11.5 percent of all observations and 3.2–11.4 percent of treated observations. The corresponding seed benchmarks range from 1.3 to 4.6 percent and from 1.8 to 3.5 percent.

Native midpoint cutoffs can route evaluation observations differently after a nonlinear recoding. Table 4 removes this difference by assigning identical dense ranks to equivalent columns on the pooled observed and counterfactual array. Both dictionaries use these same ranks, the original numerical `mtry`, folds, seed, and Riesz estimates. Only the forest inputs are recoded. The orthogonal score retains its original dictionary.

Table 4: ATET comparisons with identical canonical rank coding

Sample / specification	Full	Reduced	Change	Sign changes (%)
NSW / 1	3,142.3 (1,340.2)	3,066.5 (1,290.1)	-75.8	3.36
NSW / 2	3,221.9 (1,410.8)	3,289.5 (1,413.2)	+67.6	2.80
PSID / 1	1,506.4 (971.5)	1,646.2 (941.2)	+139.8	2.52
PSID / 2	1,385.5 (949.7)	1,336.4 (934.9)	-49.1	2.63
CPS / 1	1,677.7 (616.3)	1,597.1 (607.0)	-80.6	6.83
CPS / 2	1,551.6 (618.1)	1,522.5 (609.7)	-29.2	8.34

Notes: ATETs and changes are in dollars, with standard errors in parentheses. Changes are reduced minus full. Both dictionaries use 1,000 trees, seed 1, the same five folds, and `mtry` equal to 8 or 10. Sign changes concern the cross-fitted outcome-regression contrast over all observations.

With canonical ranks, absolute ATET changes range from 29 to 140 dollars, and 2.5–8.3 percent of individual contrasts change sign. Aggregate signs and 95 percent significance classifications remain unchanged. Thus nonlinear midpoint routing does not account for all of the first-stage sensitivity. Column-dependent tie handling and pseudorandom draws can still differ. Neither empirical comparison measures accuracy or regret, since the true conditional effects are unknown.

Code and outputs are available in the replication repository. From its root, `python3 scripts/verify_replication.py` reconstructs the tables. The package documents full refits, recodings, retained columns, software versions, folds, and seeds.

6 Conclusion

Feature subsampling makes column allocation part of a forest's specification. The product of path depth and singleton inclusion probability determines the weight placed on a CATE component in the forest's limiting target. Growing information-equivalent representations can therefore yield opposite CATE signs and treatment rules, although every fixed finite representation covered by the theorem recovers under the stated conditions. In the Monte Carlo, class sampling lowers treatment-rule regret while increasing CATE squared error. The Econometrica replication finds movement in individual first stages alongside stable aggregate ATET conclusions. Class sampling prevents sign reversals caused solely by split-equivalent recodings on the certified array without discarding certified split actions.

A Proofs

Proof of Theorem 1. All probability bounds for the endpoints are uniform over $a \in \mathcal{C}_a$. Write $\mathcal{D}_n$ for the training data and the fitted nuisance functions, including their training randomization. Uniform boundedness, the fixed two-dimensional rectangle class, and relative VC Bernstein bounds give

$$r_n = \sqrt{\frac{\log(B_n s_n)}{k_n}} + \frac{\log(B_n s_n)}{k_n} + \|\widehat{m} - m_a\|_\infty = o_p(1).$$

These bounds hold simultaneously for the root splitting and estimation samples and all their rectangular restrictions. They therefore cover both ordinary and failure trees, even though the latter use conditional transition laws. Increasing recodings do not enlarge this rectangle class. The score identity and its perturbation bound are

$$\mathbb{E}(\Gamma_i \mid X_i = x) = \tau_a(x), \qquad \max_i |\widehat{\Gamma}_i - \Gamma_i| \le 2\|\widehat{m} - m_a\|_\infty.$$

Combining the concentration bounds with (5) yields

$$\sup_{C,j,t} \left| \widehat{\mathcal{Q}}^L_C(j,t) - \mathcal{Q}_C(j,t) \right| = o_p(1), \tag{14}$$

where $\mathcal{Q}_C$ is the population score gain.

First consider a node before coordinate j has been used. Its range in that coordinate is still $[0,1]$, and independence removes the other component from the split contrast. Direct calculation gives

$$\mathcal{Q}_C(j,t) = \begin{cases} a_j^2 t/(1-t), & 0 < t \le 1/2, \\ a_j^2 (1-t)/t, & 1/2 \le t < 1. \end{cases} \tag{15}$$

The unique maximizer is $1/2$, with gain a_j^2. Positivity of $\underline{a}$ and the coefficient gap imply $a_2^2 - a_1^2 \ge 2\underline{a}\Delta_a > 0$. At nodes with at least $C_0 k_n$ splitting observations, cuts approaching $1/2$ are size-admissible and balanced on the concentration event. Maximization and (14) thus imply

$$\max_{j,F} |\widehat{t}_{j,F} - 1/2| = o_p(1), \tag{16}$$

where F ranges over first-use nodes above this size band in the ordinary and failure trees.

The continuation condition now supplies the depth bound. It is used here in addition to balance. If $n_d(x)$ is the splitting-sample size on a path, then, whenever a split is made,

$$\beta n_d(x) \le n_{d+1}(x) \le (1-\beta) n_d(x).$$

The root size is proportional to s_n, no path stops above $C_0 k_n$, and every child has at least k_n observations. Consequently there are constants $0 < c_\beta < C_\beta < \infty$ and an integer K, independent of n, a, h, such that, with probability tending to one,

$$c_\beta H_n \le J_{h,n}(X), J^0_{h,n}(X) \le C_\beta H_n. \tag{17}$$

Here $J_{h,n}$ is the number of candidate draws on an ordinary path, and $J^0_{h,n}$ is defined when $\omega_n < 1$. At most K draws occur after the path enters the band $n_d < C_0 k_n$, including a final unsuccessful search. Reached leaves have splitting sizes between k_n and $C_0 k_n$. Relative concentration gives their population masses of order k_n/s_n and honest estimation counts of order k_n. In particular,

$$\max_{b,L} \left| \frac{1}{N^{\mathrm{est}}_{b,n}(L)} \sum_{i \in I^{\mathrm{est}}_{b,n} : X_i \in L} \widehat{\Gamma}_i - \mathbb{E}\{\tau_a(X) \mid X \in L\} \right| = o_p(1). \tag{18}$$

The maximum includes both representations. The counts are positive on this event.

We next control the approximation after first use. Choose a deterministic $\delta_n \downarrow 0$ so that (16) is at most δ_n with probability tending to one. For each coordinate, its first-use nodes above the terminal band form an antichain. Label only their children by the sign suggested by the side of the cut, and retain this label in their descendants. The total population mass receiving a wrong label is

$$\sum_F P_X(F) |\widehat{t}_{j,F} - 1/2| \le \delta_n,$$

where the sum is over these above-band first-use nodes. For a descendant C, let $q(C)$ be the conditional fraction with the wrong label. Then

$$\sup_t \mathcal{Q}_C(j,t) \le 4\overline{a}^2 q(C), \qquad \int_C |\mathbb{E}(s_j \mid C) - s_j(x)|\, dP_X(x) \le 4 P_X(C) q(C).$$

These bounds also apply after further splits. For $u_n = \sqrt{\delta_n}$, select the first descendant on each path for which $q(C) > u_n$. Those descendants are disjoint, so their combined mass is at most δ_n/u_n. Outside this vanishing set, a previously used coordinate has gain at most $4\overline{a}^2 u_n$ at every subsequent node. An offered unused coordinate has gain at least $\underline{a}^2 - o_p(1)$ above the terminal band and therefore wins. Only the root can involve competition between two unused coordinates.

The repeated coordinate is offered with probability at least $1/2$ at each draw. Together with (17), the preceding ordering argument bounds its probability of remaining unused above the terminal band by

$$2^{-\lfloor c_\beta H_n \rfloor + K + 1} + o(1) = o(1).$$

For the singleton, the probability of first use only in that band is also $o(1)$. If $\omega_n \le H_n^{-1/2}$, it is at most $K\omega_n$. Otherwise, avoiding an earlier use has probability at most $\exp\{-\omega_n(\lfloor c_\beta H_n \rfloor - K - 1)\} + o(1)$. The error terms here include the exceptional target mass just bounded.

Let $D_{h,n}(x)$ indicate first use of the singleton in a generic tree, and let $\check{T}_{h,n}^{\mathrm{tree}}$ be that tree's honest score prediction. If $D_{h,n}(x) = 0$, its leaf has no restriction in the singleton coordinate, whose conditional mean is exactly zero. For a first use above the terminal band, localization and the wrong-label bound control the integrated approximation error. A late first use adds at most twice its probability to the integrated error for the sign component, which vanishes by the preceding bound. The same bound applies to the repeated component. Thus

$$\mathbb{E}_\Theta\left[\left\| \check{T}_{h,n}^{\mathrm{tree}} - D_{h,n}\tau_a - (1 - D_{h,n}) a_h s_h \right\|_{L^1(P_X)} \middle| \mathcal{D}_n \right] = o_p(1). \tag{19}$$

This is an approximation after first use, not an assertion that every leaf is exactly pure.

Suppose now that $\omega_n < 1$. Conditional on $Q_{M_n} = q$, the singleton is included with probability q/M_n, so its unconditional inclusion probability at a node is ω_n. On the locally conditioned failure tree, write $\rho_{h,n,t}^0(x)$ for the probability that the ordinary transition at the current node chooses the singleton as its splitting coordinate. It satisfies $0 \le \rho_{h,n,t}^0 \le \omega_n$. After the root and above the terminal band, this probability equals ω_n outside the exceptional paths. With $\Lambda_{h,n}^0 = \sum_{t \le J_{h,n}^0} \rho_{h,n,t}^0$, we obtain

$$\left| \Lambda_{h,n}^0 - J_{h,n}^0 \omega_n \right| \le (K+1)\omega_n \tag{20}$$

outside a set of joint training, failure-tree, and test-point probability tending to zero.

Node locality ensures that conditioning transitions on other branches leaves the transition kernels on the target path unchanged. At each stage, factor the probability of avoiding a singleton split from the transition kernel conditional on that event. Multiplication along the path and integration over the conditional kernels give the exact identity

$$u_{h,n}(x) := \mathbb{P}_\Theta\{D_{h,n}(x) = 0 \mid \mathcal{D}_n\} = \mathbb{E}_{\Theta^0}\left[\prod_{t=1}^{J_{h,n}^0(x)} (1 - \rho_{h,n,t}^0(x)) \middle| \mathcal{D}_n \right]. \tag{21}$$

The kernels are well-defined because their conditioning probabilities are at least $1-\omega_n > 0$. This identity concerns no first use throughout the path, exactly the event defining $D_{h,n} = 0$.

If $\mathcal{I}_n \to 0$, then $\omega_n \to 0$ and $\Lambda^0_{h,n} \to_p 0$ by (17) and (20). The product converges to one since its deficit is at most $\Lambda^0_{h,n}$. If $\mathcal{I}_n \to \infty$ and $\omega_n < 1$, the cumulative hazard diverges and

$$\prod_t (1-\rho^0_{h,n,t}) \leq e^{-\Lambda^0_{h,n}} \longrightarrow_p 0.$$

For indices with $\omega_n = 1$, both coordinates are always offered. The stronger one wins at the root and the other at the next interior node, outside the same exceptional set. Thus recovery also holds on these indices, without a failure-tree conditioning on a null event.

For the intermediate case, fix $a \in \mathcal{C}_a$. The condition $\mathcal{I}_n \to \kappa \in (0,\infty)$ implies $\omega_n \to 0$. Equations (7) and (20) yield $\Lambda^0_{h,n} \to_p \kappa d_h(a)$. Moreover,

$$0 \leq -\log \prod_t (1-\rho^0_{h,n,t}) - \Lambda^0_{h,n} \leq \frac{\omega_n \Lambda^0_{h,n}}{1-\omega_n} \longrightarrow_p 0.$$

The product therefore converges to $e^{-\kappa d_h(a)}$. All products are bounded. Joint convergence with an independent test point implies convergence of $u_{h,n}$ in $L^1(P_X)$, in probability over $\mathcal{D}_n$, to one, zero, or $e^{-\kappa d_h(a)}$ in the respective regimes.

Finally, conditional on $\mathcal{D}_n$, independent tree groups of bounded size and bounded score predictions give

$$\left\| \check{T}_{h,n} - \mathbb{E}_\Theta(\check{T}^{\text{tree}}_{h,n} \mid \mathcal{D}_n) \right\|_{L^1(P_X)} = O_p(B_n^{-1/2}) = o_p(1).$$

Combine this with (19), the three limits for $u_{h,n}$, and the prediction transfer in (5). This proves (8), (9), and (10).

A strictly increasing recoding maps a threshold t to $\varphi(t)$ and preserves $\varphi(x) \leq \varphi(t)$ exactly when $x \leq t$. All cells and concentration bounds above are expressed in the original coordinates. Their constants therefore do not depend on the number or form of the increasing recodings. $\square$

Proof of Corollary 1. In the selection regime, Markov's inequality and (8) give, for $g = 1, 2$,

$$\mathbb{P}_X\{\operatorname{sgn}(\widehat{T}_{g,n}) \neq s_g\} \leq \frac{2}{\underline{a}} \|\widehat{T}_{g,n} - a_g s_g\|_{L^1(P_X)} = o_p(1).$$

The independent signs s_1 and s_2 disagree with probability one half. On that event, $|\tau_a| = a_2 - a_1 \geq \Delta_a$. A second application of Markov's inequality shows that both fitted magnitudes exceed $\underline{a}/2$ outside a set of P_X-mass $o_p(1)$. This proves (11) and the stated separation from zero.

For any rule π,

$$\mathrm{Reg}_a(\pi) = \mathbb{E}[|\tau_a(X)|\mathbf{1}\{\pi(X) \neq \pi_a^*(X)\}] .$$

Since $a_2 > a_1$, $\pi_a^* = \mathbf{1}\{s_2 = 1\}$, so the limit of $\widehat{\pi}_{2,n}$ has zero regret. For the other representation,

$$\mathrm{Reg}_a(\mathbf{1}\{s_1 = 1\}) = \mathbb{E}[\tau_a\{\mathbf{1}\{s_2 = 1\} - \mathbf{1}\{s_1 = 1\}\}] = \frac{a_2 - a_1}{2}.$$

Boundedness and classification convergence transfer these values to the estimated rules, proving (12). $\square$

Proof of Corollary 2. The capped-Poisson rule satisfies

$$\mathbb{E}Q_M = \xi_M + e^{-\xi_M} - \mathbb{E}(N_M - M)_+.$$

For fixed $\xi_M = \xi$, the final term vanishes as $M \to \infty$, giving $\omega_n \sim (\xi + e^{-\xi})/M_n$. If $\xi_M \to \infty$ and $\xi_M/M \to 0$, Poisson concentration gives $\mathbb{E}Q_M/\xi_M \to 1$. Under the package default, $\xi_M/\sqrt{M} \to 1$, so $\omega_n \sim M_n^{-1/2}$. The two boundaries follow by substituting these expressions into $\mathcal{I}_n = H_n\omega_n$. For fixed finite M, $\omega_n > 0$ is constant and $H_n \to \infty$. $\square$

Proof of Proposition 1. Couple the two representations with the common class-indexed randomness in the proposition. At a common node they draw the same classes, evaluate the same certified actions, and apply the same scores and tie order. Equal certificates give identical unordered children, while canonical orientation sends every point in $\mathcal{X}_{\mathrm{cert}}$ to the same child. Induction gives identical tree partitions on the certified array. Leaf observations and forest weights therefore coincide. Nuisance fits coincide because they are held fixed or computed from the same canonical matrix with common randomness. The same inputs, settings, and randomization then give identical stopping decisions, predictions, variance calculations, rankings, and policy ties. $\square$

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